# 8 Model Inadequacy

*All Models are Wrong.*
—Leonard Smith, 2006[1]

## 8.1 Introduction

All models are wrong. The simplest physical system, a pendulum, if started with a particular initial condition and let go free, will soon diverge from the trajectory you calculate from the simple model of the pendulum that we all learned as school children. The easiest way to solve the pendulum is for small perturbations, where the small-angle approximation makes the calculation simpler. Such boring dynamics. Drive a pendulum such that it can reach the unstable fixed point standing upside down, and the trajectories can be fully chaotic and thoroughly unpredictable.

So if a simple pendulum model is difficult to use to track the motion of the physical device, can we anticipate that our neuronal, or cortical models, will be anything but abysmal in reflecting reality?

The answer is yes, and no.

The harsh reality is that if you really wanted to use a model of a pendulum to track a real pendulum, you would need to fit parameters as well as variables. Friction must be included to account for damping. And this dual-estimation problem of variables and parameters is one that we have already dealt with in chapter 5. Since our model of a pendulum is a *good* one, if we incorporate it into our nonlinear Kalman framework, it will do much better than an empirical model might at short-term prediction of the dynamics. This is one hallmark of a good model for our purposes—that it can be used in our recursive predictor-controller framework to accomplish accurate dynamical prediction.

But this is a book about neural control. Certainly, not only are the Hodgkin-Huxley equations, Fitzhugh-Nagumo equations, and Wilson-Cowan equations wrong, but the network

1. Second International Seizure Prediction Meeting, Bethesda, 2006.

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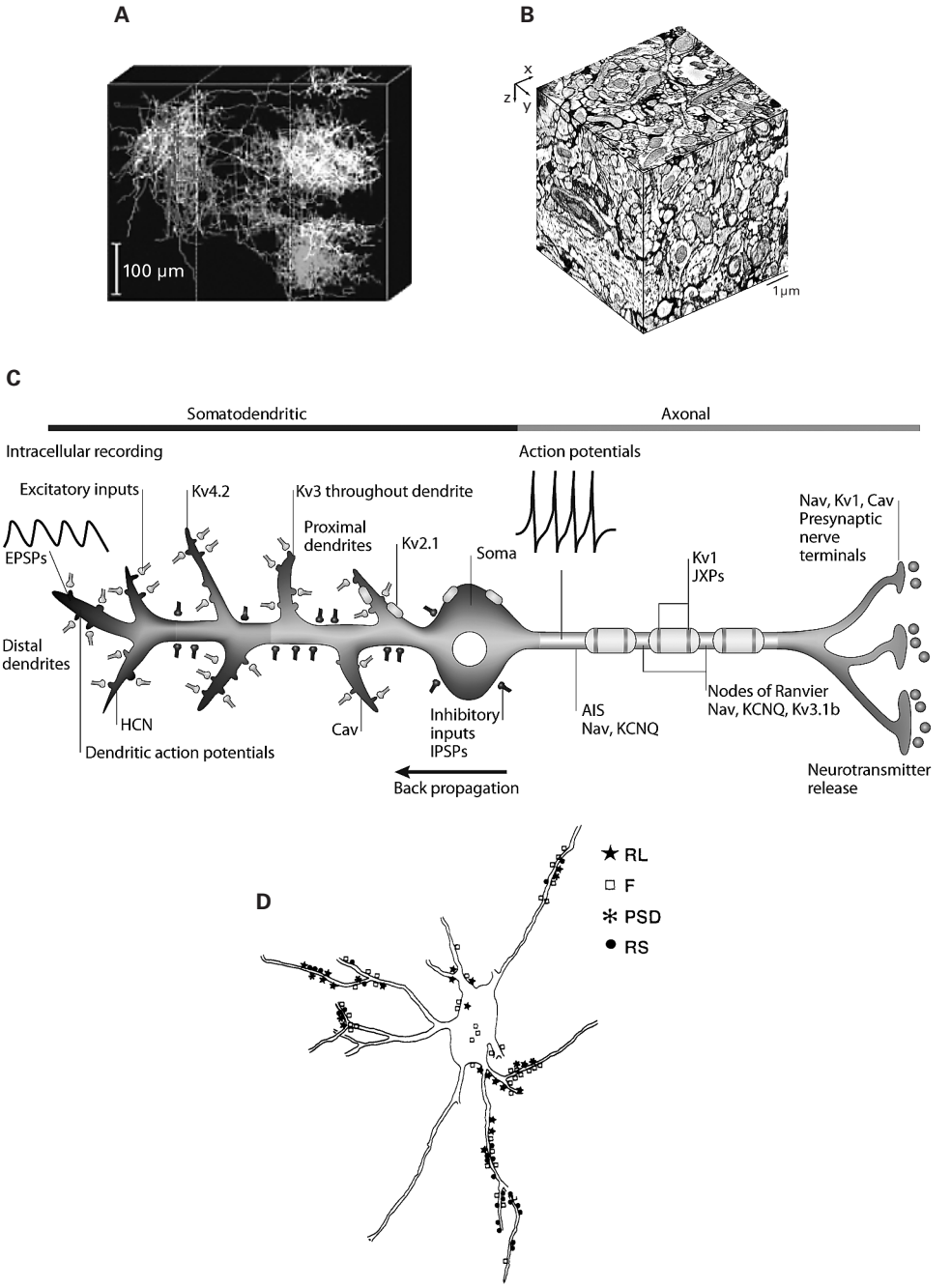



**Figure 8.1 (plate 14)**
(Caption on facing page)

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topologies we used to account for network dynamics in chapter 6 are wrong. In fact, we know almost nothing about the true network topologies of any small piece of mammalian brain. There have been heroic efforts to define connection topologies electrically cell-pair by cell-pair [MW84, MW86]. Such issues have spawned intense research into what many call the *connectome* problem, and many researchers are intensively trying to fill this gap by, for example, attempting to define the exhaustive connectivity of cubic millimeter volumes of brain using electron microscopic and machine learning algorithms [BD06, Seu09].

The complexity of single neurons is mind-boggling (figure 8.1A). The complexity of networks of neurons is both mind-boggling and mind-generating (figure 8.1B). Viewing the morphologies from figure 8.1 (plate 14) brings up a limits of knowledge question. When we needed superb models, such as of aerospace vehicles, NASA or Boeing were able to pour resources into creating models of spacecraft or airframes that were so accurate that their model-based control systems functioned superbly. But rockets and airplanes are simple compared with the neurons in figure 8.1. Not only are these mammalian neurons complicated by their topology (figures 8.1A, B), but within each neuron, the distribution of the dynamical ion channels [LJ06] (figure 8.1C) and ligand-based receptors [LHJ95] (figure 8.1D) are very inhomogenous. Would any amount of finite research effort permit the full modeling of the features in Figure 8.1, panels A, C, and D, into panels B? No one can say at present. But skepticism over the accuracy of our large-scale biophysical models of brain [Mar06] is warranted at the present time. And if such models were to be verifiably accurate in their details, fitting their variable values and parameters to data would prove intractable with current techniques. The latter statement is diplomatic. One must worry that we may face fundamental limits to knowledge in encountering the extraordinary complexity of the brain [Sco95].

All of this is not intended to instill despondency and early termination of the activity of reading this book. But it emphasizes the paramount importance of addressing *model inadequacy*. And it underscores the need for the methods presented in this book in dealing with the ever-growing complexity of computational neuroscience models.

## 8.2 The Philosophy of Model Inadequacy

All of the control theoretic frameworks so far discussed—linear, extended, and ensemble Kalman filtering—focused on *noise* as a source of error. They did not implicitly formalize

**Figure 8.1 (plate 14)**
Complexity of neurons and their networks. A shows the reconstruction of six individual interneurons in the hippocampus, whose dendrites are shown in red, and axons in yellow. Reproduced from [GDR+05] with permission. B shows an electron micrograph reconstruction of a 350-$\mu$m reconstruction of a small cube of mammalian neuropil into which one must envision the neurons from A are embedded. Reproduced from [BD06] with permission. C shows an inhomogenous distribution of voltage-gated ion channels in a neuron. Reproduced from [LJ06] with permission. D shows an inhomogenous distribution of synapse type on a neuron. Reproduced from [LHJ95] with permission.

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that the models were bad. Nevertheless, every control engineer recognizes that the uncertainty in such frameworks lump into the noise terms the uncertainty in the models themselves.

But even the very act of sampling systems introduces model error. Natural and manmade systems generally express their dynamics in continuous time. Most digitized implementations of control frameworks sample such continuous systems discretely, and then iterate a discrete model to project such dynamics forward in time. *All discrete predictor-corrector implementations of continuous systems are by definition approximations to continuous truth.*

Certainly with respect to the nervous system, we typically sample data at time scales relatively long with respect to the underlying dynamics. While collecting data, we are not iterating the models. While iterating the models, we are not sampling data. Although clearly making a case for the parallel implementation of such roles, the point is to underscore that such iterative and discrete computation misses a lot.

Another fundamental issue is that models tend to be formed at particular temporal and spatial scales of the hierarchical organization of a natural system. For neural dynamics, we generally assume that we can focus on cellular membrane voltage and spike dynamics, without delving down to the level of quantum indeterminacy or quarks [BK86]. In terms of time, we think that milliseconds count, rather than femtoseconds.[2] Our models *cannot* account for scales of dynamics below the resolution of the model. We typically lump all such fine scale errors into the noise terms in Kalman filtering applications. But this ubiquitous scale dichotomy between nature and model is at its essence another ubiquitous source of model inadequacy.

We will, in chapter 12, explicitly model lower scales into neuronal models, and show examples of their necessity. But one can always ask how the inadequacy of the representation of even lower scales diminishes the veracity of our models. And the more scales we add, the more complex a tracking model becomes. In the end, we will need to trade off model complexity against model tracking accuracy; there will always be a point at which adding more complexity, even valid complexity, results in a worse model fit. Such a fact is, however, very good news for our hopes to run such neuronal tracking models in real time.

## 8.3 The Mapping Paradigm—Initial Conditions

The first technical approach we will examine focuses on initial conditions of model iteration. A consequence of starting an imperfect model out with a given set of initial conditions is that it will map these starting points to the wrong place. So a practical approach is to change the initial conditions in order to achieve a more perfect trajectory.

2. In certain neuronal applications, such as dynamic clamp [DBNW07], or in waveform spike sorting [MPIQ09], the temporal scale of several tens of microseconds is commonly employed.

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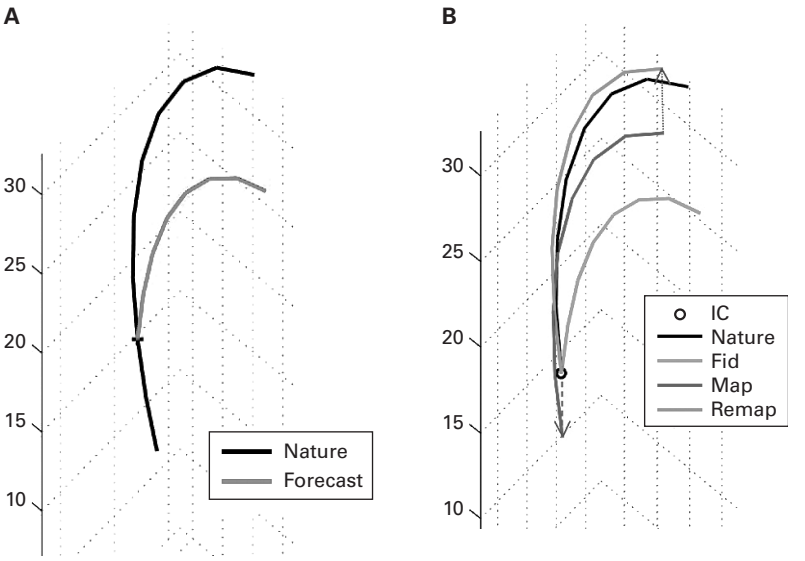



**Figure 8.2 (plate 15)**
A illustrates an imperfect model trajectory. B shows mapping strategy to bias the initial condition, IC, in order to improve the trajectory. Fid indicates the fidelity paradigm. The bias applied to the initial condition in mapping (Map) is corrected following iteration (Remap). Reproduced from [TP07] with permission.

An important subtlety here is that, faced with an imperfect model, we ordinarily consider that the first thing to do is to improve the model. The next thing to do is to get the initial conditions closer to nature. But bringing initial conditions closer to nature might worsen the results when the model is flawed. *The goal is to get the trajectory to end closer to the truth, not to get the initial conditions closer to the truth*. Toth and Peña argued in 2007 [TP07], that fidelity to the truth in initial conditions in fact is applicable only when the model is perfect. Perhaps only in highly constrained scientific or engineering experiments might this be reasonable.

A schematic of the trajectory produced by an imperfect model is illustrated in figure 8.2A (plate 15).

*Attractors* in dynamical systems theory are the asymptotic set of points to which all trajectories are, well, attracted. The physical issue involved with figure 8.2A is that the difference in dynamics between nature and the imperfect model implies that the attractor of nature is *different* from the attractor of the imperfect model. The *model drift* encountered then occurs because the initial condition, albeit close to nature, was not close to the equivalent position in the model system.

The fix for this deviant initial condition is elegantly simple: Map the natural initial condition close to the equivalent model initial condition, iterate the model, and then remap the

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model trajectory result back to the natural attractor coordinates. The idea is schematized with mapping vectors (magenta arrows) in figure 8.2B. Such a mapping vector is the asymptotic difference between nature and the model, and is not at all appropriately mapped by assuming that expanding the random errors of uncertainty will appropriately capture this mapping. A simple unoriented expansion of covariance inflation (expansion of the sigma points) would capture the mapping, but at the expense of almost pure uncertainty in essentially every direction but the mapping.

Toth and colleagues create a sample system by (of course, since they are working on meteorology) modifying the Lorenz 1963 model that we introduced in chapter 5 in the following way. They take the Lorenz equations from equations (5.4),

$$\dot{x} = -\sigma x + \sigma y$$

$$\dot{y} = -xz + rx - y$$

$$\dot{z} = xy - bz$$

translate the attractor in the $z$ direction with $z = z + 2.5$, and change the $\sigma$ variable from 10 to 9 to alter the dynamics. They keep $r = 28$ and $b = 8/3$ for both models.

One of the very useful aspects of the mapping paradigm is that one can look at historical data (whether weather or neurons), or run models for a while, and derive the mapping vector as the difference in the mean values of the two attractors: nature versus model.

Figure 8.2 was actually plotted from these two Lorenz systems. In figure 8.3A are shown the sequential steps of the fidelity paradigm (blue) and the remapped predictions (green), and the prediction errors for such sequential steps for these procedures are shown in figure 8.3B (see plate 16).

Note in figure 8.3B that if you keep using any of these models, the forecast errors eventually asymptote to the same values. The parameters used in both of the Lorenz systems were in the chaotic range. You can shadow a true trajectory of a chaotic system for only so long—the dynamics through sensitive dependence of initial conditions destroys information, and eventually you are just guessing the mean value of the system. This mathematical observation fits with our efforts to predict the weather. Beyond a time horizon of about two weeks, our detailed weather predictions are useless. We think that the weather is chaotic because of such observations, and it is one of the reasons why so much of the numerical meteorology literature uses chaotic systems to bench test prediction algorithms.[3] One of the characteristics of stationary nonlinear dynamical systems, even if chaotic, is that although their short-term predicability is limited, their long-term statistical averages such as mean (or dimension, or entropy) are rock steady.

3. Is this applicable to nervous systems? We seem adapted to be able to mirror and predict what our acquaintances are going to do in the short term. We generally have no idea what they will do two weeks from now. But following this analogy any further is not sensible.

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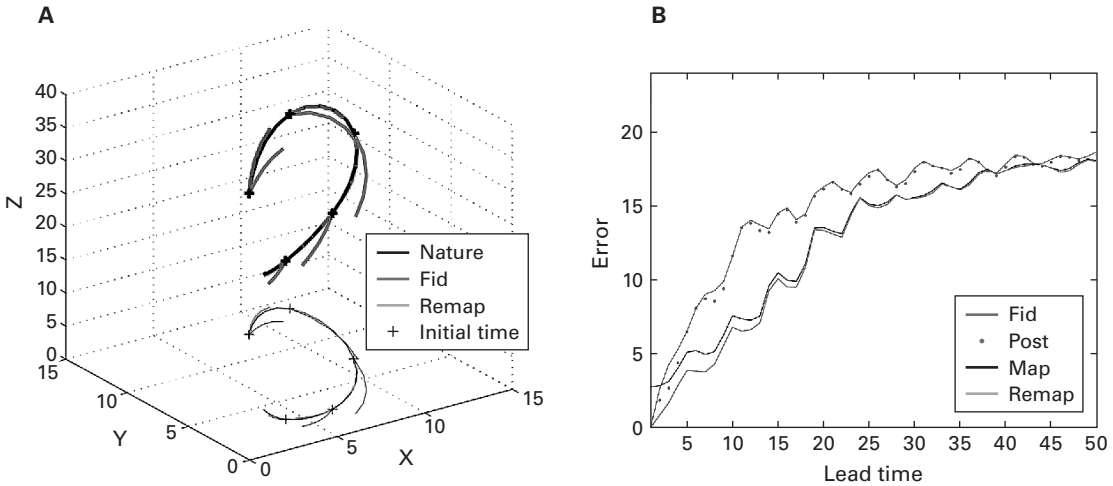



**Figure 8.3 (plate 16)**
Sequential application, A, of the mapping-remaping (green) versus the fidelity (Fid) paradigm in the two forms of the Lorenz equations as described in the text. On the right are shown the sequential errors between the two methods. Standard post-processing (Post) to remove bias. Reproduced from [TP07] with permission.

What if the natural system is not stationary with time?[4] Then the mean between nature and model, the mapping vector, will change as a function of time. The solution is to apply this mapping paradigm adaptively by keeping a running average of the offset between measurement and predicted model state, the innovations $y - \hat{y}$ (chapter 2). Averaging a set of innovations lets you estimate and update the mapping vector at some reasonable frequency with this time-dependent mean difference [TP07].

The mapping paradigm requires some cautions. Toth and colleagues show examples where they assume that they know the dimensions and variables of both nature and the model. This is not actually necessary for implementing such a mapping paradigm. You always have measurements of nature and your prediction of the state of nature, and you can always create a mapping vector by taking their *difference*. But at the time of this writing, the full implications of applying this elegant mapping vector to cases where we explicitly understand that the state space, variables, and dimension of the natural system are unknown remains unclear to the author.

Toth and colleagues avoided augmenting their state vector with model offsets. Their mapping paradigm was calculated *outside* of their Kalman filter. They did this to achieve simplicity, yet there is no reason to presume that model inadequacy lives outside of the state

4. We will define nonstationarity as time-dependent parameters. In the case where the entire model itself varies with time, one might envision *switching* models [Mur98]. But the more general problem of changing attractors has no general solution.

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of a system. Very likely, a given model fits nature better at some times than others—that is, the model errors are correlated with state. Toth and colleagues were aware of an earlier work that attempted to address these issues, but its complexity and difficulties seemed to be substantial. This deep work brings us to the heart of a powerful abstraction of model inadequacy, which we focus on next.

## 8.4 The Transformation Paradigm

[W]e treat the forecast model as a black box, that does not yield the true time evolution of the atmosphere, and we attempt to use this black box in conjunction with observations to account for model bias in the state estimation.
—Baek et al., 2006 [BHK$^+$06]

Baek and colleagues [BHK$^+$06] sought to explicitly incorporate model inadequacy into an ensemble Kalman filter framework. They focused on model errors that have no random component, and call this offset *model bias*.

Their approach is to augment state vectors with bulk model error terms. They estimate these error terms dynamically with their ensemble Kalman filter, seeking to *parameterize* the forms of model bias.

Baek and colleagues assume that the model is a finite state representation of an infinite dimensional system. They assume that the model equations are not precisely known, and they expressly ignore natural dynamics at finer scales than the model.

Let's assume that there is a true state, denoted by $x^t$, for time $n$, of a natural system whose dynamics, $F^t$, can be described as

$$x^t_{n+1} = F^t(x^t_n)$$

Let's further assume that there is a model system and state, denoted by $x^m$, whose dynamics $F^m$ are governed by

$$x^m_{n+1} = F^m(x^m_n)$$

Almost always, the dimensions of $x^t_n > x^m_n$, but for now we will assume that they are the same. There are three general bias models:

### Bias Model I

Let's assume, in the fidelity paradigm of Toth et al. [TP07], that ideally we place the initial condition of the model very close to the initial condition of nature, so that $x^m_{n-1} \approx x^t_{n-1}$. Then the *bias* is

$$b^t_n = F^t(x^t_{n-1}) - F^m(x^t_{n-1})$$

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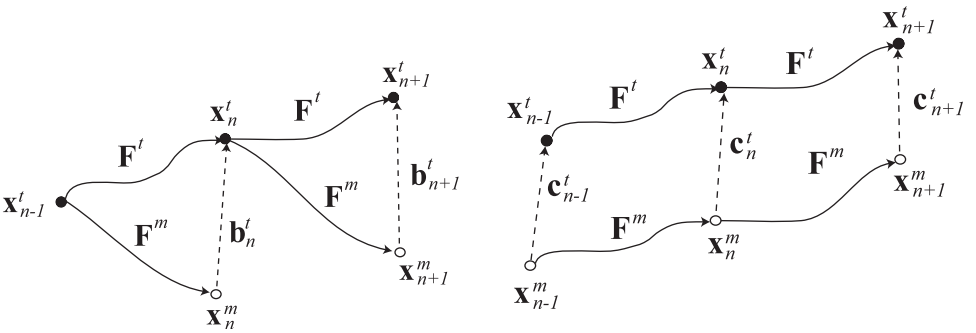



**Figure 8.4**
Bias model I is shown in A. Because of the differences in true, $F^t$ versus model $F^m$ dynamics, the bias, $b$, must be accounted for. B shows the contrasting geometry from bias model II. Reproduced from [BHK+06] with permission.

as illustrated in figure 8.4. If the bias is constant in time, its dynamics are expressed as

$$x^t_n = F^m(x^t_{n-1}) + b^t_n$$

$$b^t_n = b^t_{n-1}$$

The constant model for $b$ is one that we have encountered previously, in chapters 2 and 5, and we can treat it as a parameter with trivial dynamics in an ensemble Kalman filter. This model assumes that the best forecast is produced when the fidelity paradigm is operative, that is, you can place the initial condition very close to that of nature, $x^m_{n-1} \approx x^t_{n-1}$. Indeed, we assume that following correction through $x^t_n = F^m(x^t_{n-1}) + b^t_n$, our observation function is

$$y = A(x^m_n + b^t_n) + \varepsilon_n$$

where $\varepsilon$ represents random measurement error.

Bias model I, by itself, is not as good as the mapping paradigm described by Toth and Peña [TP07], which leads us to bias model II.

### Bias Model II

Let's assume that our model system has a different attractor from nature. We expect that this is always true. Put your best and most perfect initial condition measured from fidelity with nature, and no simple additive drift such as bias model I can account for the discrepancy.

We seek a transformation of coordinates from nature to the model, as

$$c^t_n = F^t(x^t_{n-1}) - F^m(x^m_{n-1})$$

$$= F^t(x^t_{n-1}) - F^m(x^t_{n-1} - c^t_{n-1})$$

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The geometry of bias model II is brilliantly illustrated in figure 8.4B reproduced from [BHK+06]. The system dynamics can now be approximated, if we assume a constant model for $c$:

$$x_n^t = F^m(x_{n-1}^t - c_{n-1}^t) + c_n^t$$

$$c_n^t = c_{n-1}^t$$

Assuming a constant $c$ is equivalent to [TP07], but in general, one can employ nontrivial dynamics on $c$, as discussed in detail in [BHK+06]. As before, we assume that our observation function is now

$$y = A(x_n^m + c_n^t) + \varepsilon_n$$

It is now a logical step to consider bias model III.

**Bias Model III**

In bias model III, we combine bias models I and II to produce, for static models $b$ and $c$,

$$x_n^t = F^m(x_{n-1}^t - c_{n-1}^t) + b_n^t + c_n^t$$

$$b_n^t = b_{n-1}^t$$

$$c_n^t = c_{n-1}^t$$

**Parameterizing Model Inadequacy**

We now create vectors of parameters that correspond to the model bias schemes just described.

Assume that the evolution of state $x$, $dx/dt$ represents a series of $N$ ordinary differential equations that prescribe the evolution of system state. Then the most general form would be

$$\frac{dx}{dt} = F(x + c) + b$$

where

$$b = [b_1, b_2, \ldots, b_N]^T$$

$$c = [c_1, c_2, \ldots, c_N]^T$$

where $T$ indicates transpose. We now augment the state vector, $v$, to be used in the ensemble Kalman filter, with

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$$v = [x, b, c]^T$$

Finally, Baek and colleagues [BHK+06] employ covariance inflation as we used in chapter 2, which they find can stabilize the Kalman filter in the presence of model error.[5]

Figure 8.5A–C illustrates the results of numerical simulation of the Lorenz 1998 model [LA98] subjected to model biases I, II, and III. Interestingly, the use of type III bias parameters always gives the best results, whether the true model bias is type I, II, or III. One reason to understand this is, for instance, in the case of type I bias. In figure 8.5D, note that using type III parameterization, rather than type I parameterization, decreases the distance from the true trajectory for true type I bias.

Using the augmented state vector $v$ leads to a covariance matrix for the errors that expands from $x^2$, from chapter 5, to $(2x)^2$ for bias model II, and $(3x)^2$ for bias model III. Such augmentation dramatically increases the computational overhead when Cholesky decompositions of such a matrix are required in the calculation of sigma points.

The convergence or *settling* time for such algorithms can be large. Baek and colleagues demonstrated that this settling time is strongly influenced by the magnitude of the covariance inflation parameter. Another finding was that if they placed spatial diffusion constraints on the rate at which the model bias parameters could change, this improved the settling time. For all of the results in figure 8.5, the tuning of covariance inflation is critical to accuracy. All of this tuning was performed exhaustively offline. It's important to note in figure 8.5 that, to some degree, increasing covariance inflation can compensate for a less adequate bias model.

Another issue that the author raises is whether Baek's insertion of the model error parameters within the augmented state vector is the best place for them. The ensemble Kalman filter will tend to find mean values for the distribution of ensemble particles or sigma points. But such values might not be the best reflection of the model errors. It remains an open question whether optimizing type I, II, or III model error parameters might best be performed in the manner of Toth and Peña [TP07], outside of the Kalman filter. Such optimization might be far more computationally efficient, since the covariance matrices would be substantially smaller.

The complexity of the calculations in Baek et al. [BHK+06] increased dramatically for type I or type II parameterization. But the complexity was tremendously increased for type III. One would, if possible, want to pick just type I or type II, but the results show that type III, if the covariance inflation is tuned properly, is always numerically best.

Baek and colleagues never addressed how one might approach tuning covariance inflation in real time. We will address this next.

5. In their implementation, their covariance inflation is proportional to the trace of the state covariance matrix times the identity matrix, rather than by an arbitrary small constant times the identity matrix as we employed in chapter 5. Their formulation provides a degree of adaptability as the trace of the covariance matrix changes dynamically with time.

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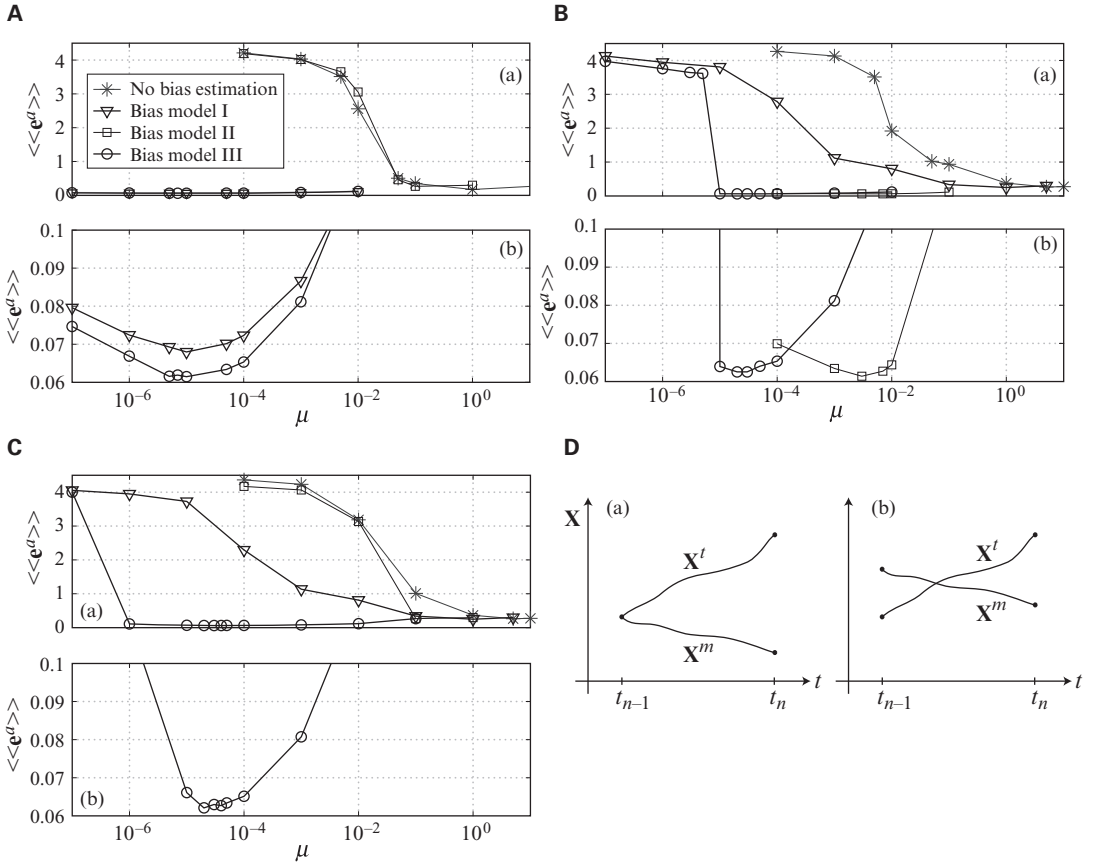



**Figure 8.5**
Employing the three model error parameterization schemes for type I model bias (A), type II model bias (B) and type III model bias (C) The ordinates are forecast error, while the abscissa for A–C are covariance inflation. Note that type III model bias, for appropriate tuning of covariance inflation, always gives the best results. One reason for this is illustrated in D, where in true type I bias, the overall error is less when model parameter c is used. Reproduced from [BHK$^+$06] with permission.

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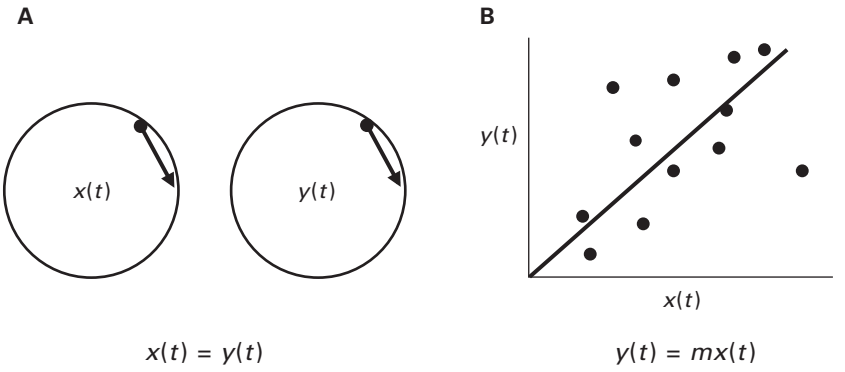



**Figure 8.6**
Schematic of two systems in synchrony in A, and inferring synchrony through correlation in B.

## 8.5 Generalized Synchrony

These findings lend credibility to the suggestion that a synchronization principle is also fundamental to the relationship between the brain and the external world,
—Duane et al., 2006 [DTW06]

Consider for a moment what a Kalman observer system is doing. It is a dynamical system that, if perfect, would emulate the system it hopes to track. It was recognized several years ago that control systems seek to *synchronize* with the natural system. In an airplane cockpit, the model of the airframe dynamics seeks to perfectly synchronize with what the aircraft is actually doing.

Synchrony implies that two systems, $x(t)$ and $y(t)$, do the same thing at the same time: $x(t) = y(t)$ (figure 8.6A). For identical systems, synchronization can be exact. For non-identical systems, they can come close, but never exactly synchronize.

Synchrony is traditionally inferred from correlation (in time) or coherence (in frequency). Figure 8.6B shows that, for noisy data, we often try to demonstrate that a linear functional relationship relates two systems to each other when they are synchronizing. For linear systems (which do not exist), this is reasonable.

It came as a surprise to the dynamics community when Lou Pecora and Tom Carroll demonstrated in 1990 that chaotic systems could synchronize [PC90]. In figure 8.7, we show one of their examples for two sets of chaotic equations. Although the state of the system on the right starts far away from the drive system on the left, it rapidly synchronizes to it. There are some serious subtleties to chaotic synchronization. Both systems remain chaotic. But if you perturb the driven system, there are no longer expanding directions. All small perturbations stably return to the same trajectory, which is the same as the drive system that enslaves it. This is an important condition for all synchronization—it is stable to perturbation. Another way to envision this is that when synchronized, such driven systems

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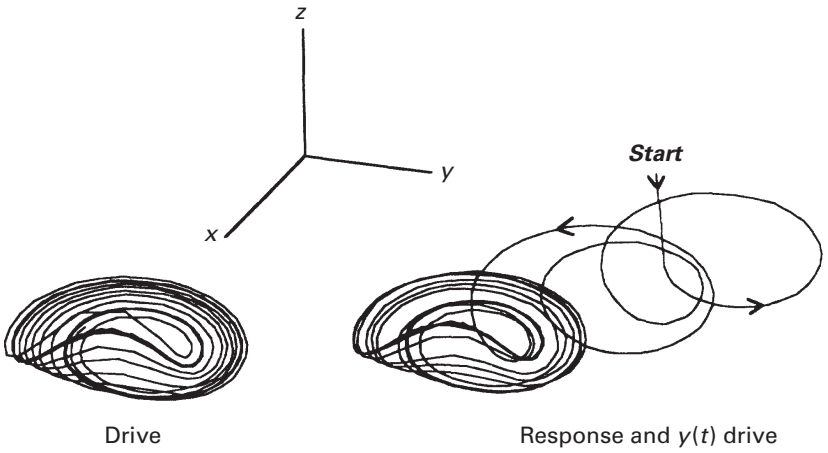



**Figure 8.7**
Chaotic synchronization of Rössler equations. Reproduced from [PC90] with permission.

*forget* their initial conditions and adopt the dynamics of the driving system. In neuroscience applications, we rarely apply stability tests to infer synchrony, relying on correlation in time, although there are exceptions [FGS03].

What if the systems were not identical? We never expect systems to be identical. And in this book, we never anticipate that they have linear dynamics. In 1995, Nikolai Rulkov and colleagues [RSTA95] defined what they termed *generalized synchronization* for dissimilar coupled nonlinear systems. Figure 8.8 shows examples of the difference, as coupling ($C$) is increased, between identical and nonidentical nonlinear systems as they synchronize.

When we defined linear correlation in figure 8.6B, we defined a *functional relationship* between two systems. For nonidentical systems, a straight line is no longer adequate to describe the relationship. The function is nonlinear.

Defining metrics to tell when such dissimilar nonlinear systems were or were not in synchrony really depends on whether you can map from one system's state to another. If there was a 1:1 correspondence, then mathematically a homeomorphism might be present. Such 1:1 correspondence could be inferred through testing for *continuity*. Furthermore, if the functional relationship preserved the orientation within such continuity functions, there might be differentiability of the function—a *diffeomorphism* between the two systems. A detailed set of mathematical and statistical tests emerged to quantify generalized synchrony [PCH95]. Another way to quantify such synchronization was through mutual prediction, and this was applied to neural networks and single neurons in mammalian spinal cord circuitry in [SSC$^+$96].

Duane and colleagues [DTW06] proposed that data assimilation through Kalman filtering can be envisioned as a synchronization problem. Without a controller, the coupling between

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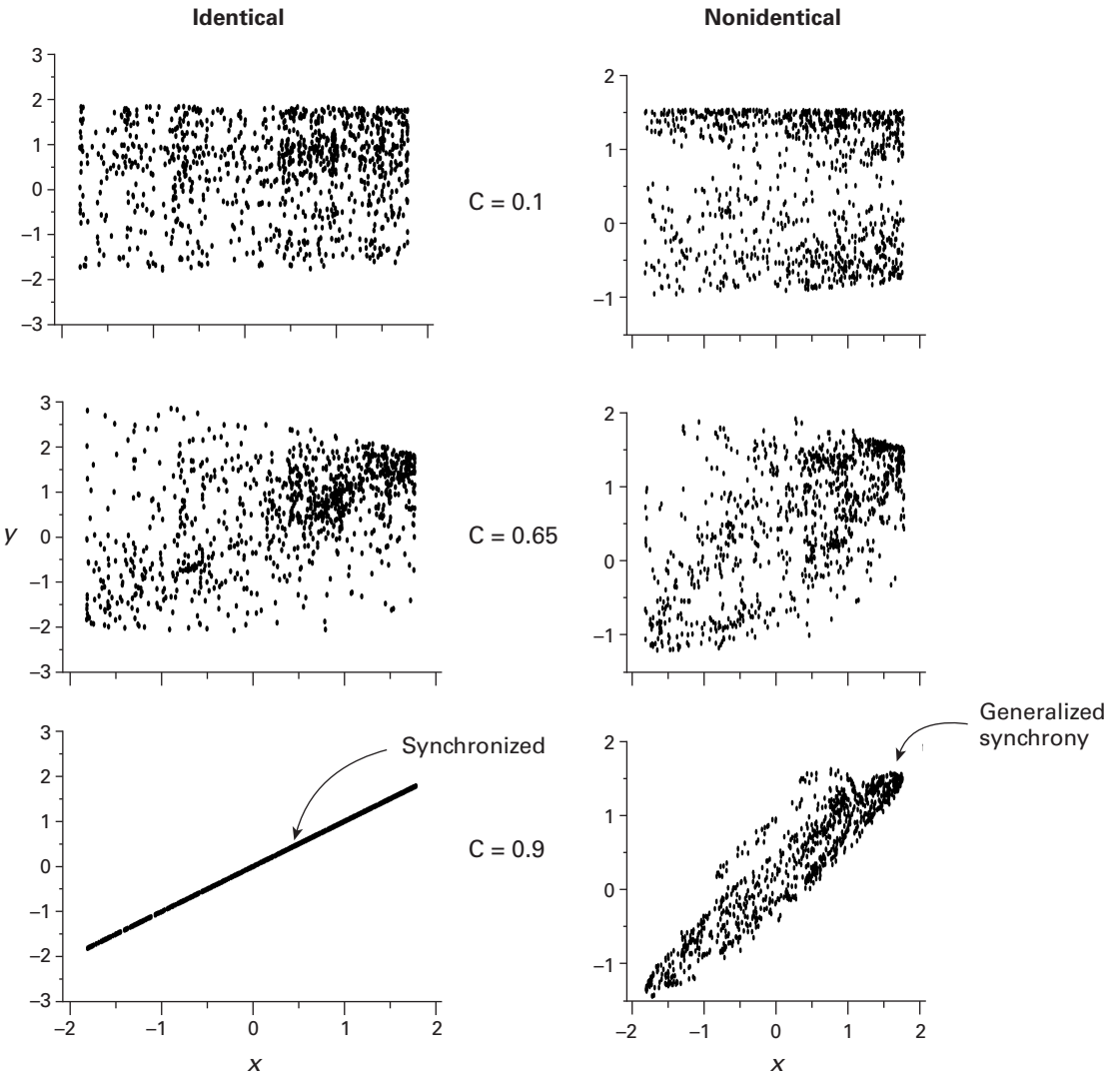



**Figure 8.8**
Contrast between identical and nonidentical nonlinear systems as a coupling parameter, $C$, is increased. Identical (left) and generalized (right) synchronization patterns are illustrated. Reproduced from [SSC+96] with permission.

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nature and the observing model is one way, and the situation is similar to the coupling for nonlinear systems described in [PC90] or [RSTA95]. They also speculated that some of the psychology of Carl Jung on *synchronicities* [JP55a] might fit within such an observing synchrony framework (we defer further discussion of Jung until chapter 13).

They examined a simple example of tracking a system where the attractor switched from one set of Lorenz 1963 equations to another

$$\left.\begin{aligned} \frac{dx}{dt} &= \sigma(y-x) \\ \frac{dy}{dt} &= rx - y - xz \\ \frac{dz}{dt} &= xy - bz \end{aligned}\right\} \rightarrow \left\{\begin{aligned} \frac{dx}{dt} &= zy - bx \\ \frac{dy}{dt} &= rx - y - xz \\ \frac{dz}{dt} &= \sigma(y-z) \end{aligned}\right.$$

In figure 8.9, they illustrate an experiment where they use a Kalman filter to track the above switching system. It does a reasonable job, except that at the point of switching

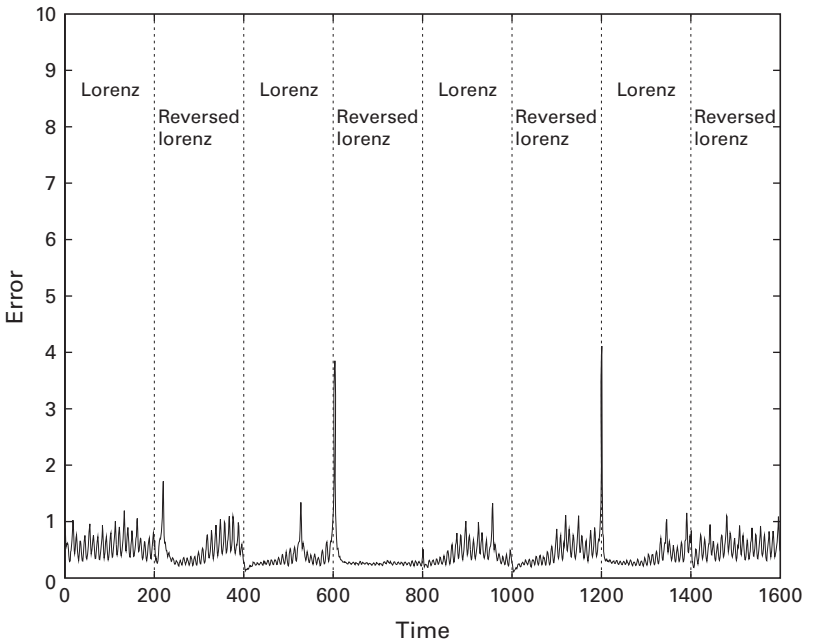



**Figure 8.9**
Kalman filter loses track of a system during rapid changes in dynamics. During such time there are tracking catastrophes. Increasing covariance can help mitigate against this. Reproduced from [DTW06].

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(dotted lines), there are tracking *catastrophes* as the filter looses track of the system, and it takes finite time to reacquire the trajectory. These desynchronization bursts are seen when the trajectory of the coupled system blows off of the synchronization manifold that had previously defined generalized synchrony.

One way to decrease the effect of these blowouts is to increase covariance inflation so that the tracking ensemble is broad enough to not lose the trajectory. And a strategy to deal with anticipating such blowouts is considered next.

## 8.6 Data Assimilation as Synchronization of Truth and Model

It seems unlikely that a group of undergraduate students gathered during their summer vacations under a grant to foster research would produce a seminal work at the cutting edge of control theory, but the work of Yang and colleagues appears to be that rare event [YBL+06].

Using a coupling function, $G$, of the difference between a variable, $\alpha$, and its observed measure, $\alpha_{obs}$, was termed *nudging* by Hoke and Anthes [HA76]. Yang and colleagues [YBL+06] proposed nudging the Lorenz 1963 equations as follows:

$$\frac{dx_s}{dt} = \sigma(y_s - x_s) + \delta_x k(\bar{x} - x_s)$$

$$\frac{dy_s}{dt} = r x_s - y_s - x_s z_s + \delta_y k(\bar{y} - y_s)$$

$$\frac{dz_s}{dt} = x_s y_s - b z_s + \delta_z k(\bar{z} - z_s)$$

where $s$ refers to the *slave* system being driven by nature, $k$ is a coupling constant (the inverse of which is a time scale of the coupling), $\bar{x}$, $\bar{y}$, and $\bar{z}$ are the observations of nature, and the $\delta_{x,y,z}$ are indicator functions equal to 1 when the nudging is in the direction of a given variable, and zero otherwise. Simple nudging through the $x$ variables of two Lorenz systems was shown by Pecora and Carroll [PC90] (see figure 8.7).

Yang and colleagues [YBL+06] wished to observe a noisy Lorenz system, and sought to synchronize to it using nudging. So they set up their coupling to use all combinations of variables: $x$, $y$, $xz$, $yz$, $xy$, and $xyz$.[6] In figure 8.10, we show the errors of such coupling. There are some notable things to comment on. First, coupling through all variables is not necessarily as good as coupling through only one or two variables. The efficacy of coupling depends on coupling strength. These issues open a new dimension to Kalman observers—that picking the variables to observe, and the strength of the coupling gain on the observable

6. Coupling through $z$ alone is often unsuccessful in the literature for the Lorenz 1963 equations.

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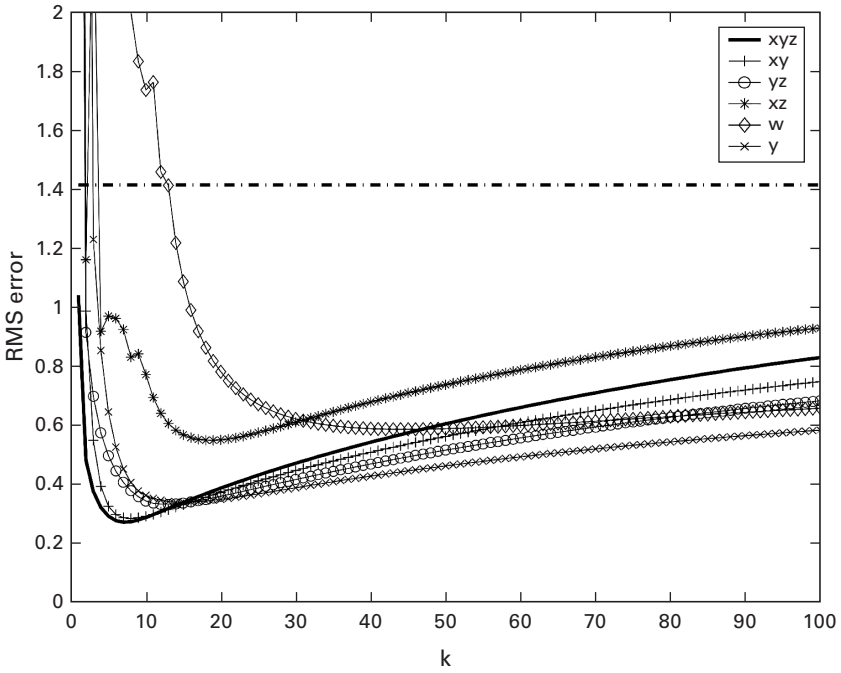



**Figure 8.10**
Synchrony error versus coupling strength for a variety of coupling variables in the Lorenz 1963 equations. Reproduced from [YBL+06] with permission.

variables, are factors that will affect the ability of the observer system to track a natural system. Intriguingly, Yang and colleagues found that if the sampling interval was lengthened, increasing the coupling strength would help compensate for this.

The next finding of Yang and colleagues was that as the synchronization coupling strength was lowered below the value that would synchronize the systems, intermittent bursts of desynchronization (tracking catastrophes) would be observed. Such *intermittency* in dynamics is well described. It can be a consequence of the system dwelling close to but not asymptotically on an attractor, when the dramatic excursions are termed *blowout bifurcations* [OS94]. Or, especially attractive to our purposes, such excursions can be found in attracting systems when noise is intruduced, and this has been termed *bubbling*, especially in the case of synchronizing systems [ABS94].

### Where the Wind Blows

We take a brief excursion to examine *bred vectors*. If we took weather data, and measured, say, wind speed velocity from a grid of points above our heads, we can iterate the vector of wind velocities forward through our favorite weather model and see what the forecast is. We could also perturb this initial condition, making sigma points as in chapter 2, and

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push each of the ensemble points through the forecast model. Let's say that our ensemble of $k$ initial points were spherical in state space about the initial condition. Then, if the atmosphere operator were a matrix, matrix times circle equals ellipse (chapter 7), and we have an ellipsoid. Perform a singular value decomposition on the covariance matrix $B^T B$ of the ensemble of $k$ bred vectors, $B$, and you can ask which of the expanding directions is most significant [PHK+01].

The largest expanding direction(s) are where the initial state was most sensitive to perturbation. You can define a *bred vector dimension* $\psi$ from

$$\psi(\sigma_1, \ldots, \sigma_n) = \frac{\left(\sum_{i=1}^{k} \sigma_i\right)^2}{\left(\sum_{i=1}^{k} \sigma_i^2\right)}$$

where if each column of $B$ were normalized to 1, then $\sum_{i=1}^{k} \sigma_i^2 = k$. The key to this computation is that the number of singular values, $\sigma_i$, that are significantly larger than zero will dictate an effective dimension between 1 and k. So let's look at what this implies for the winds of the world in figure 8.11 (plate 17). San Diego, California, is easy to predict. It's the same every day. Miami, Florida, is (with respect to wind speed) chaotic. So let's breed these vectors in a control context.

### From Wind to Control

We will breed with the slave system (the observer), since the natural system can be considered unknown. This is the case for nearly every control process we will encounter with the nervous system. The bred vector, calculated from sigma points surrounding the state estimate applied to the model system, predicts where the system will be growing. One can use nudging in the direction of the bred vectors to correct this expected blowout/bubbling. In figure 8.12A we present the calculations from [YBL+06], showing how nudging from the bred vectors gives better synchrony with a true system than simple nudging. Note that the substantial excursions from synchrony, the peaks in panel A, occur as intermittent desynchronous bursts. But bred vectors give us far more than that. Look at panel B in figure 8.12. The color coding indicates the magnitude of the singular values of the largest bred vector, superimposed on the Lorenz $x$ variable. Blue is small, green is medium, and red is the largest in magnitude. The large-magnitude bred vectors tell you when the system is about to blow out of synchrony in panel (see plate 18).

All of this is well and good if you know the equations for nature. You don't. But you do know the equations of your observer model.

If you couple (unidirectionally) a natural system to a model system, at sufficient levels of coupling, you may develop a functional relationship, $\psi$, between the drive and response system, as schematized in figure 8.13. For unidirectional coupling, the response system

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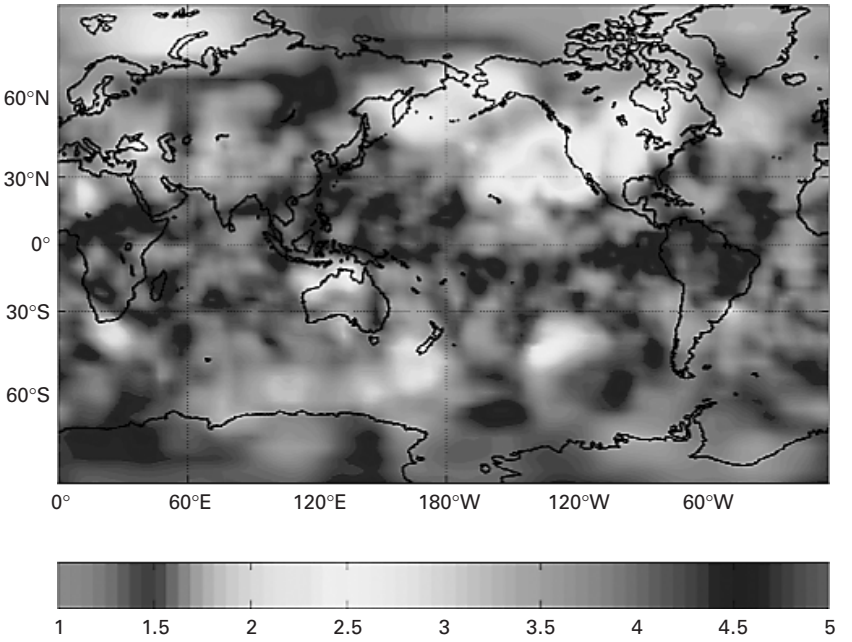



**Figure 8.11 (plate 17)**
Bred vector dimension for wind speed across the world. Reproduced from [PHK+01] with permission.

always gives you information on the drive system, so you have some drive system predictability from observing the slave system before it is enslaved. Once it is sufficiently coupled, the drive system can enslave the slave system. This means that a stable functional relationship can develop. At this point, the existence of the function means that you not only can predict the drive system, but the drive system can predict the response system. The prediction becomes bidirectional. If you were using a method such as continuity [FSGS01] to test the existence of a functional relationship, you would see that for a given set of points, $\epsilon$, in the response system, they all came from a given place, $\delta$, in the drive system [PCH95, NPS04]. Our ability to detect such functions always has limits. In real life, the function will always have a certain uncertainty or thickness, and we can define the limits to the detection of generalized synchrony in such settings [SBJ+02]. Nevertheless, our ability to detect synchronization (and the absence thereof) in settings of increasing nonlinearity and noise is improving, and further discussion can be found in [NPS04, SSKW05, NCPS06].

Key in this picture is that the stability of the functional relationship means that the slave system must *forget its initial conditions*. So by analogy, if you started two copies of the slave system from different initial conditions, and coupled them strongly enough to a single drive system, you would find that they synchronized *to each other*. Since other than the

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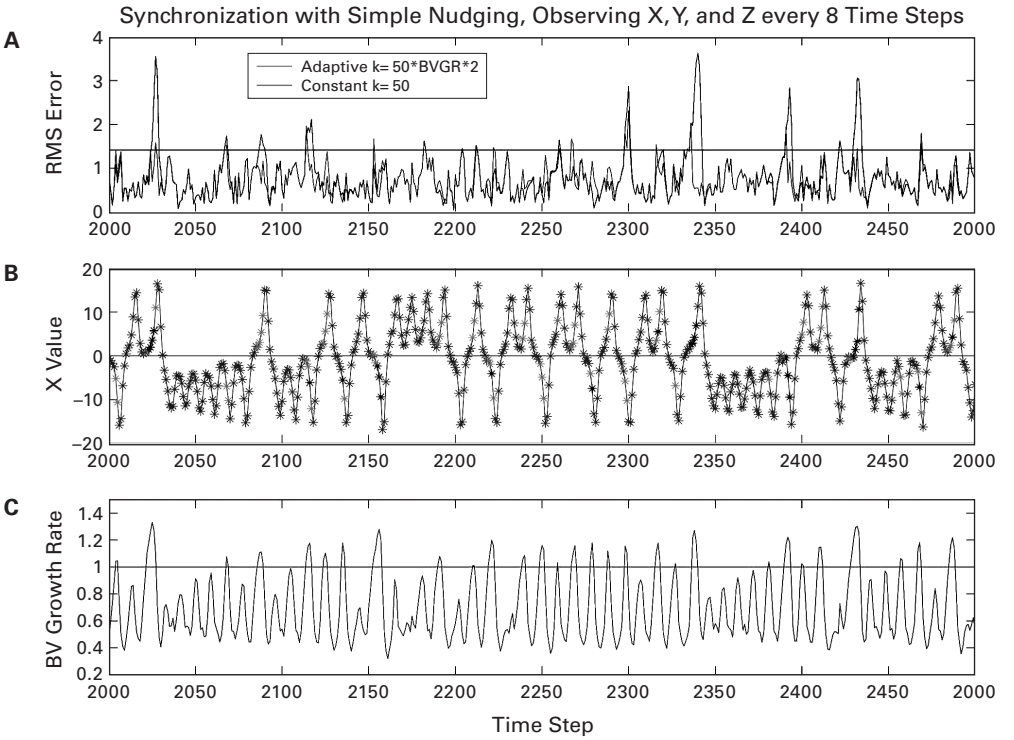



**Figure 8.12 (plate 18)**
Nudging synchrony control for Lorenz 1963 equations using simple nudging along the axes of the original variables (blue), versus along the bred vector directions (red), shown in panel A. Panel B shows the Lorenz $x$ in the slave system, along with the color-coded value of the magnitude of the singular value for the bred vector with blue being small, green being medium, and red being large in amplitude. Panel C shows the logarithmic bred vector growth rate, each averaged over 8 time steps. This growth rate is predictive of impending desynchronization. Reproduced from [YBL+06] with permission.

initial conditions, the slave systems you know, and can be identical, they will identically synchronize as schematized in figure 8.14.

As you might imagine, the conditions to see paired slave or *auxiliary* systems synchronize are not quite as strong as between the driving system and the slave systems. In 1996, Pyragas [Pyr96] defined the use of auxiliary systems to infer the possibility of generalized synchronization. He defined a *weak* and *strong* case for generalized synchronization, and considered the identical synchronization of the auxiliary systems as an expression of weak synchronization. Nevertheless, seeing such synchrony is a necessary, if not sufficient condition, for the existence of generalized synchronization between a natural system and these auxiliary systems. The situation is schematized in figure 8.15. Interestingly, Pyragas also examined the case of one completely different system (Rossler) driving two identical auxiliary systems (Lorenz). The onset of generalized synchronization can be seen at sufficient

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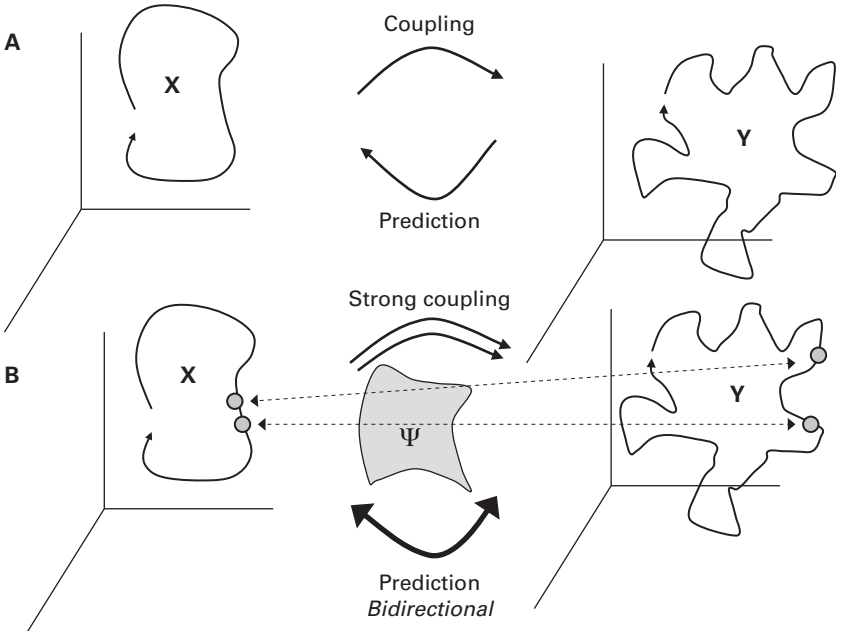



**Figure 8.13**
A shows a drive system, X, driving a response system, Y. The coupling goes from left to right, and the prediction goes from right to left for weak coupling. In B, as coupling is increased, a functional relationship develops, $\psi$, after which predictability is bidirectional.

levels of coupling in the synchronization of the auxiliary systems. We have also examined the stochastic synchronization of auxiliary systems; they can be observed to synchronize from a purely stochastic drive [SSC$^+$96]. This stretches the bounds of what we envision as the functional relationship between slave and drive. The slave system has information from the stochastic drive system, but cannot predict the stochastic drive system because it is random. In contrast, at sufficient levels of coupling, the slave system is predictable from the drive systems [SSC$^+$96]. Such a situation does establish a formal relationship between drive and response systems, and the auxiliary systems do tell you *something* about the uncharacterized drive system. This situation is further discussed in [YBL$^+$06]. You don't have to know everything about a natural system to have your tracking be useful.

Now back to control.

If one sets up a Lorenz system driver, and a set of matched copies of auxiliary systems with different parameters from the drive and with different initial conditions from each other, one can observe an example of such auxiliary systems synchronizing (figure 8.16, plate 19).

Note the time needed to observe synchronization. This points to a fundamental difference between synchrony and typical Kalman frameworks. Kalman filtering predicts the best guess

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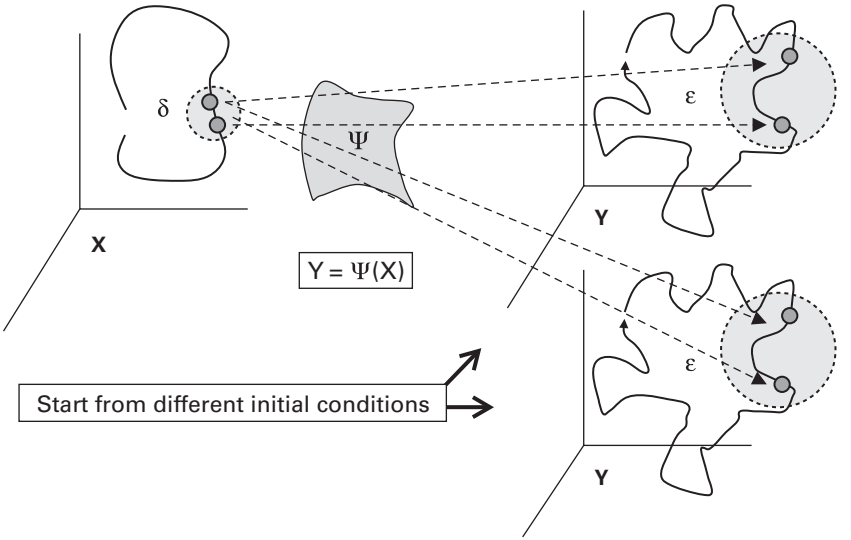



**Figure 8.14**
Example of driving two copies of response systems. Starting from different initial conditions in the response systems, they will identically synchronize with each other under conditions when generalized synchronization with the drive system develops in each. If the drive system dynamics is unknown, we may be able to observe the auxiliary response systems synchronize with each other.

at the present state. A bred vector approach tells you what is about to happen a short time into the future. Kalman filters are a bit of a *nowcast* [DTW06].

And finally, Yang and colleagues [YBL$^+$06] offer a prescription for adaptive adjustment of observation models in anticipation of an increase in forecast errors. If one were either to calculate the amplitude of the bred vector singular values (figure 8.12B), or average the values for several recent time steps (figure 8.12C), such an increase in local instability of the synchronization could predict the desynchronization bursts shown in figure 8.12A. The results pave the path for what we have not had up until now—a way to adaptively tune *covariance inflation* on the fly. Yang and colleagues also suggest tuning the coupling strength or gain on the fly. All of this can be proposed based on calculating bred vectors on the fly using auxiliary systems. One can envision spawning auxiliary systems, with random perturbations, at periodic intervals.

But how to inflate covariance with respect to bred vectors remains an open issue. In Yang et al. [YBL$^+$06], the focus is on the direction of greatest expansion. In certain dynamical systems, actually in probably *every* real natural system, the number of expanding and contracting directions will not be constant as a system evolves. The mathematical term for this is non-hyperbolic. In such systems, the *thin* contracting directions of a set of initial conditions may cause the propagated model trajectories to miss the real trajectories entirely.

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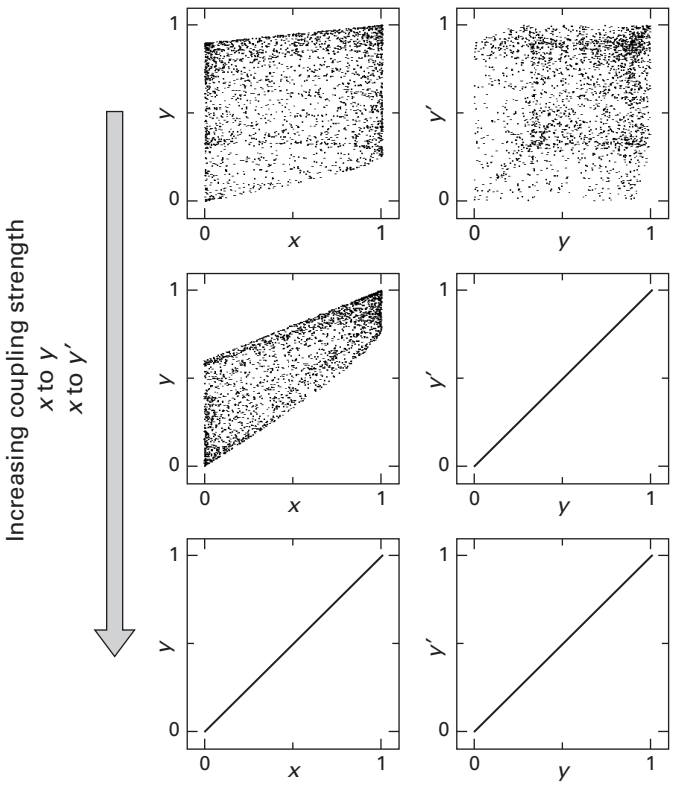



**Figure 8.15**
As coupling is increased from top to bottom, a drive system, $x$, will synchronize with a response system, $y$. In addition, copies of the response system, $y$ and $y'$, will synchronize to each other, but at a lower coupling than that required to see synchrony between the driver and response systems. Reproduced from [Pyr96] with permission.

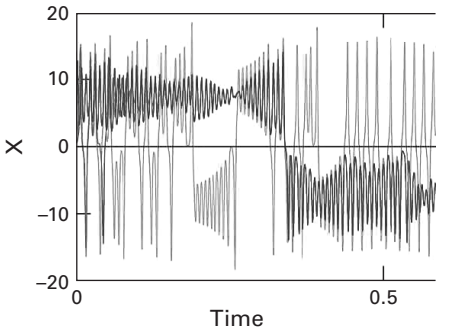



**Figure 8.16 (plate 19)**
Two identical auxiliary slave systems, in pink and blue, driven by a dissimilar drive system in green. The auxiliary systems synchronize rapidly to each other. Modified from [YBL+06] with permission.

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One might need to inflate such initial conditions along the thin or contracting dimensions [DY06]. Reconcilling these two tales of inflation will require additional research, but they both point to the same conclusion. Use enough anticipatory inflation to keep close to real trajectories.

And if auxiliary systems were run for only one time step, what are they equivalent to? They are our sigma points from previous chapters—each is a perturbed copy of our slave system. In addition to calculating the mean of the iterated sigma points, one can use the singular value decomposition of the covariation of the iterated sigma points to estimate whether the system is going to be prone to impending instability or not. If so, then add more covariance inflation to try to hold onto the system for the expected wild ride to come. There is more data in the ensemble Kalman filter than has been previously appreciated.

**Fusion - Synchronous Nudging and Kalman Filtering**

In a fusion of the concepts discussed in this chapter, one can also add a control system on an observer system to help it synchronize to a natural system. The natural system would be uncontrolled in this pure data assimilation scenario. Why do this? Because this is a way to adaptively estimate both state and parameters of the natural system dynamically over time.

Discussions of such adaptive control of the synchoronization of an observer system can be found in [Hua06] and [DYK07].

If we consider a dynamical system

$$\dot{x} = f(x)$$

Where $x$ is an augmented state of varibales and parameters. Then following [Hua06], we can design an adaptive feedback controller for an observer system

$$\dot{y} = f(y)$$

by adding a control term, $u$, as

$$\dot{y} = f(y) + u$$

where the control law

$$u = -k(y - x)$$

and coupling strength for positive $k$ can be adaptively estimated as

$$\dot{k} = \lambda(y - x)^2$$

for small $\lambda$.[7]

7. It is probably best to split the variables, $x$, from the parameters, $p$, in such systems as $\dot{x} = f(x, p)$, and for the observing system, as $\dot{y} = f(y, q) + u$. One then develops a separate evolution term for the parameters as $\dot{q} = -\beta(y - x)f(y)$, where the $\beta$ are a vector of arbitrary constants [DWC09].

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If we knew $x$, then for low noise and uncertainty in model and system, such an adaptive nudging scheme would serve to synchronize the observer to the natural system. Of course, this is a chapter about ubiquitous uncertainty and, of course, we never know $x$.

It was [DWC09] who suggested fusing the above adaptive nudging scheme with an unscented Kalman filter estimation of $x$, $\hat{x}$, as

$$\dot{y} = f(y) - k(y - \hat{x}) \tag{8.1}$$

Deng and colleagues [DWC09] go further to apply the approach in (8.1) to both the Fitzhugh-Nagumo and the Hindmarsh-Rose neuronal models. In this application, they assumed compete knowledge of the underlying equations, and demonstrated that this *combined* approach (adaptive nudging and unscented Kalman filter) to parameter estimation was extremely robust to added noise. The further application of their combined approach with the concepts of model uncertainty described in this chapter remain open problems begging for exploration.

## 8.7 The Consensus Set

Finally in our brief tour of model inadequacy, we will return to cortex and real data. In our implementation of the Wilson-Cowan equations in chapter 6, we assumed that each node in these equations had the same parameters. Such an assumption is unreasonable for brain, no matter how homogeneous or isotropic we imagine it to be.

In their most general form, the Wilson-Cowan equations can be written as follows, where for each $(i, j)$ grid point, we have

$$\dot{u}_{ij} = c_{ij1}u_{ij} + c_{ij2}a_{ij} + c_{ij3}\sum_{i'j'} e^{-d} H(u_{i'j'} - c_{ij4})$$

$$\dot{a}_{ij} = c_{ij5}u_{ij} + c_{ij6}a_{ij} \tag{8.2}$$

where the sum is taken over all other nodes $i'j'$ in the network, $d$ denotes the distance between grid points $ij$ and $i'j'$, and $H$ denotes the Heaviside function or a smoothed version (such as a steep sigmoid) if a continuous equation is desired. The $u$ variable represents instantaneous excitation, and the $a$ variable represents recovery or adaptation. Biologically meaningful parameters ordinarily require $c_{ij1}, c_{ij2}, c_{ij6} < 0$ and $c_{ij3}, c_{ij5} > 0$.

We will assume that the true system is heterogeneous, and vary the parameters for each node by 10%. For a $16 \times 16$ grid, we have $256 \times 6 = 1,536$ actual parameters. Fitting such a multiplicity of parameters would be unfeasible. One way to address this is to let each parameter float in the ensemble Kalman filter, and see if they will settle down and converge to optimal values that are the best local mean fits for the grid in question. In [SS09], we show that this is feasible, as shown in figure 8.17 (plate 20). But this demonstration is a

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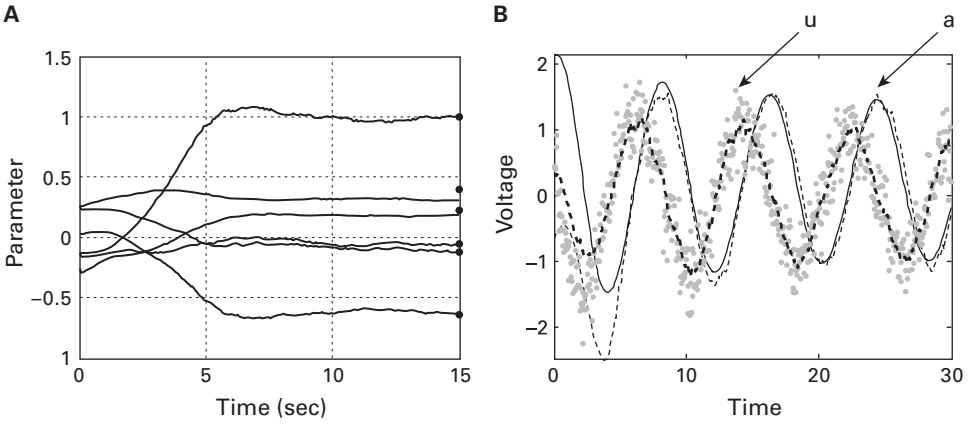



**Figure 8.17 (plate 20)**
A shows that when tracking a heterogeneous network, the ensemble Kalman filter can be used to let the parameters float and find average values that converge close to the actual mean parameter values in the network being tracked (filled circles). B shows that, from one of the grid points, the $u$ and $a$ variables are closely tracked. Reproduced from [SS09] with permission.

bit unrealistic, to say the least. We used knowledge of both $u$ and $a$, and we used the same model (with heterogeneity) to both generate the data and to assimilate it.

Before going further, we need to deal with symmetries (see also discussion in chapter 6). Symmetries in equations are important to address in any Kalman filtering assimilation of data. The reconstruction algorithm cannot distinguish between variables that are functions of each other. Consider the task of determining $a(t)$ and $c_1, \ldots, c_6$ in equations (8.2) from the knowledge of $u(t)$ alone (we drop the $ij$ subscripts for simplicity). Any experimental data $u$ that is consistent with $a(t), c_1, c_2, c_3, c_4, c_5, c_6$ is consistent with $a(t)/\gamma, c_1, \gamma c_2, c_3, c_4, c_5/\gamma, c_6$ for arbitrary nonzero $\gamma$, meaning that the dual estimation problem is underdetermined. Applying the ensemble Kalman filter directly to this problem results in parameter drift. To achieve data assimilation in this context, we must force $\gamma$ to assume a unique value.

Now we apply such symmetry-less observation equations to real data from the cortex. We will employ voltage-sensitive dye imaging data from [HTY$^+$04] (see chapter 6), and seek a consensus set in our assimilation model. The results are shown in figure 8.18 (plate 21). There are some important subtleties in these results. The consensus set is the one that predictively tracks the data the best. *It is not going to reengineer the brain and reveal the physiological parameters that correspond to our tracking model*. The parameters in figure 8.18B are physically meaningless. But the convergence of the tracking is excellent in figure 8.18C.

Simultaneous with our work in brains, came the results of Cornick and colleagues in fluids [CHO$^+$09]. They were studying Rayleigh-Bénard convection experiments. In these

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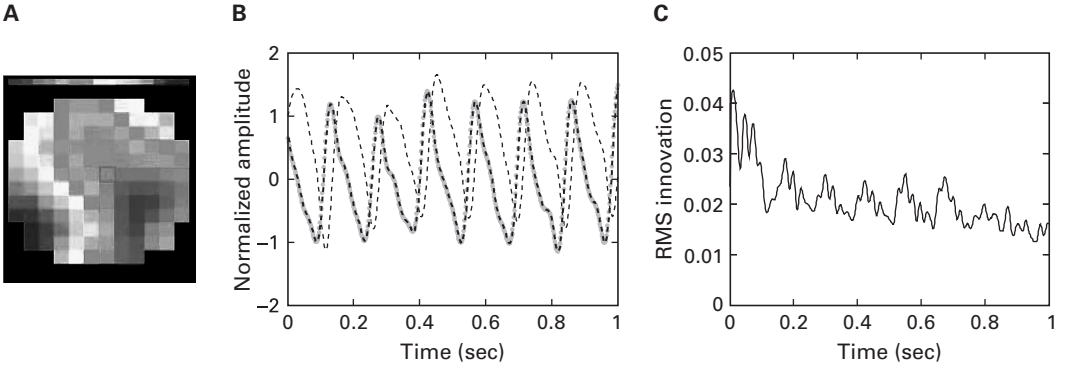



**Figure 8.18 (plate 21)**
A snapshot of voltage-sensitive dye data from a sequence of thousands of snapshots collected at 1,000 samples per second (A), the consensus set tracking of a single Wilson-Cowan node (B), and the root mean square (RMS) convergence of the tracking errors (C). Reproduced from [SS09] with permission.

experiments, a thin depth of fluid is heated uniformly from below by a plate, and convection rolls develop. The physics is described by the Navier-Stokes equations, but they used a simpler approximation to these equations, the Boussinesq equations, to assimilate the data. They employed an ensemble Kalman filter, and use a tiling of the spatial domain [OHS$^+$04] to locally approximate the system. Their results [CHO$^+$09] are shown in figure 8.19. There are several important points to note about these results for our purposes. First, in case the reader was wondering whether directly inserting data into an assimilation model might be a good shortcut to using the Bayesian assimilation of a Kalman filter, the forecast errors are always worse (figure 8.19). Use a filter. The next point is that the parameters in their tracking model that best tracked the experiment were *not* the most physical ones, but the ones that best reduced tracking error. This is exactly what we concluded above regarding the tracking of brain data [SS09].

How do you know how much to tile data? Well, if you can get away with just one tile, as we used in figure 8.18, that will increase the efficiency of calculations. The next step is to break up the spatial field to see if tiling the space will increase accuracy. The tiling scheme in [OHS$^+$04] is elegant. Obviously, the ultimate tiling would be to fit each cell of, say, a Wilson-Cowan network to assimilate each channel of data—a very computationally expensive proposition. The answer is probably straightforward. Start with one large tile, and increase the number of tiles until you hit a minimum in tracking error. The more tiles, the harder it is to fit parameters. Such tiling optimization has not, to our knowledge, been well explored in applications.

The current research that is driving the study of model inadequacy is coming primarily from branches of science where the natural systems, meteorology and fluid turbulence, are both spatially and dynamically very complex. They share complexity features with our

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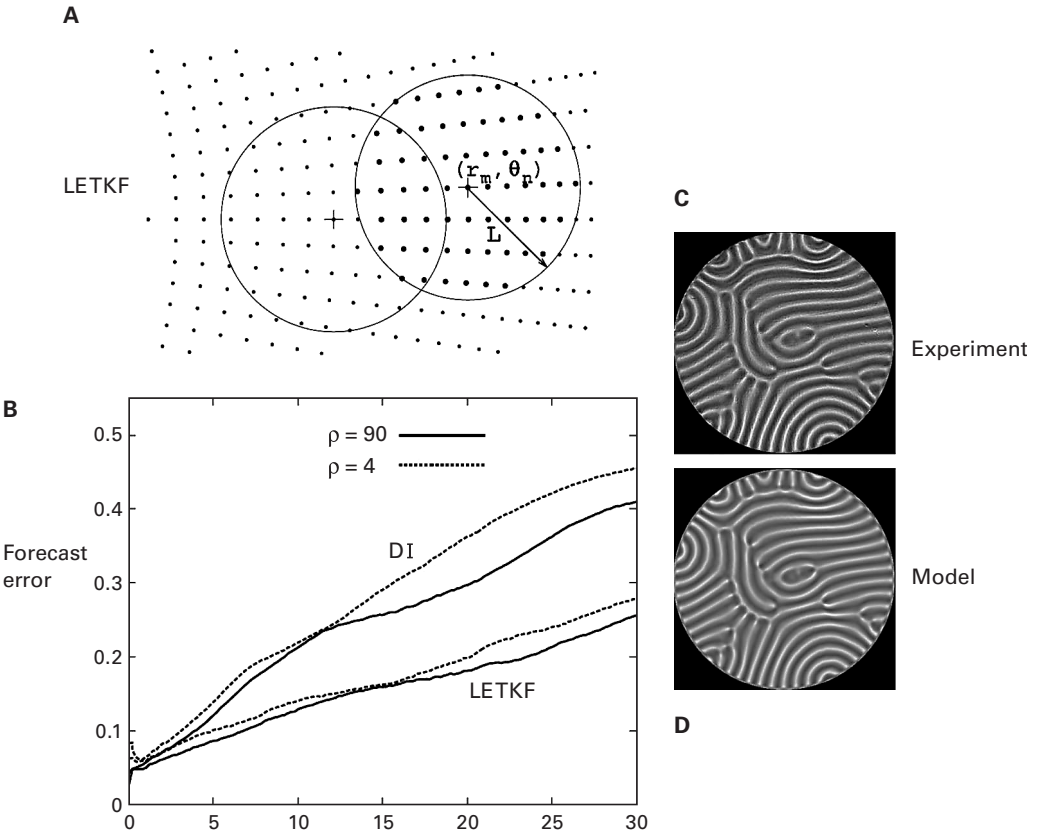



**Figure 8.19**
The local tiling of Raleigh-Bénard convection experiment (local ensemble transform Kalman filter), and forecast errors using the Boussinesq equations for data assimilation. In the forecast error plot, direct insertion of data, DI, is always worse than using the ensemble Kalman filter; p is the sparseness of sampling. Reproduced from [CHO$^+$09] with permission.

observational problems in the nervous system. We have the delightful prospect that although our colleagues in meteorology will long be stuck simply observing and forecasting weather, we should be able to use their findings to develop feedback control schemes for dynamical diseases of the brain.

## Exercises

**8.1.** Let's replicate the model from section 8.3 [TP07]. Use the Lorenz 1963 equations for your data assimilation model, setting $\sigma = 10$, $r = 28$, and $b = 8/3$.

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$$\dot{x} = -\sigma x + \sigma y$$

$$\dot{y} = -xz + rx - y$$

$$\dot{z} = xy - bz$$

Now modify the equations that you want to track by translating them in the $z$ direction with $z = z + 2.5$, and change the $\sigma$ variable from 10 to 9 to alter the dynamics. Keep everything else the same.

• You need to calculate the mapping vector. Run both models for a while, so that initial transients die down, and observe their means. It should not matter what the initial conditions are, but you might best consider using several were this a real application. For now, if you use the same initial conditions each time for both systems, that is enough for starters.

• What are the errors, the innovations $(y - \hat{y})$, in tracking for the model with and without using the mapping paradigm? How does it compare with figure 8.3B.

• Now set up the data-generating model so that the dynamics drift. You might want to change $z = z + 2.5 * t/T$, where $t$ is the time, and $T$ is the total time of the simulation, and do this *only* in the underlying system that you want to track (i.e., assume that you do not know that this drift is present when you do the tracking). Set up a mapping paradigm as before, using a long simulation to generate the mean mapping vectors. Now try calculating a running average of the innovations $(y - \hat{y})$ in order to estimate the mapping vector. Can you improve the tracking errors by doing this?

**8.2.** Now set up the same scenario as in exercise 8.1 again. Use the altered Lorenz equations as nature, and use the original Lorenz equations as your data assimilation model. Leave the time-dependent drift out for now (the $t/T$ term).

• Implement Bias Model II from section 8.4. Augment the state variable by a vector of $b$'s, one for each variable $x$, $y$, $z$ to serve as drift offsets. In principle, this will give very similar results to the mapping paradigm. Creating these uncertainty parameters is a more adaptable strategy than using a fixed mapping paradigm, but the increase in size of the covariance calculations Pxx will slow things down. Try a range of covariance inflation for several runs, and plot the mean tracking errors.

• Implement Bias Model III by further augmenting the state variable with a vector of $c$'s, one for each variable $x$, $y$, $z$ to serve as variable offsets. Now your Pxx will be 9 times larger. But the offset should fit the offset z variable well. Try a range of covariance inflation for several runs, and plot the mean tracking errors.

**8.3.** Again, set up the same scenario as in exercise 8.1. Use the altered Lorenz equations as nature, and use the original Lorenz equations as your data assimilation model. Leave the time-dependent drift out for now (the $t/T$ term). Don't apply the mapping paradigm or the bias models as in exercises 8.1 or 8.2.

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• Your sigma points are philosophically each a set of perturbed model states arranged in a spheroid. If we arrange them as columns in a matrix $B$, they are equivalent to the perturbed wind vectors of section 8.6. When you iterate them, they form an ellipsoid. Recall matrix times circle equals ellipse from chapter 7. Form a covariance matrix of such an iteration and perform a singular value decomposition as suggested in section 8.6.

• Can you plot the largest singular value along with the tracking error at each step?

• Try using the largest singular value at each step, or averaging it for several steps, to adaptively adjust covariance inflation. Can you find a strategy that decreases tracking errors?

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